

Yash Vaghela

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Education

Queen Mary University of London, MEng Mechanical Engineering Sep 2023 – Jul 2028

- **Grade:** First Class (Expected)
- **First Year Avg:** 79.3%
- **Awards:** IMechE Best Student Certificate, QMUL SEED award, Certificate of best Intercalated/QMCUR Posters ILF 2025
- **Relevant Modules:** Computational & Mathematical Modelling, Usage of CREO to design structures, Fluid Mechanics & Thermodynamics, Materials Science

Overton Grange High School, SM2 6TQ, Sutton, United Kingdom Sep 2015 – Jul 2022

- A - Levels: **Mathematics (A*)**, **Further Mathematics (A*)**, **Physics (A*)**, **Chemistry (A*)**
- GCSEs: **All 10 GCSEs passed** with **grades 6-9** achieved (Grade 9 achieved in Mathematics, Physics, Chemistry)
- Level 2 BTEC ICT: **Distinction* (D*)**
- **Passed** Extended Algebra

Experience

Member, *Queen Mary Electronics Society* Sep 2024 – Present

- Developed skills in PCB design and circuitry through hands-on projects and team collaboration
- Implemented image and object detection into the robot's sensors to enhance autonomous functionality, with the current aim to reduce camera delay by 25%
- Utilised 3D CAD modelling to design and refine mechanical components for optimal performance

IC Electronics Deputy Head, *Queen Mary Formula Student* Feb 2024 – Present

- Revamped and implemented wiring systems for the team's Formula Student car, gaining hands-on experience in automotive electronics
- Developed and modelled circuits using KiCad and RapidHarness, optimizing designs for performance and cost efficiency
- Collaborated with the team to develop the Brake System Plausibility Device (BSPD): proposed, simulated, and tested a BSPD, achieving 99% reliability in circuit operations. Optimised inertia switch circuitry reducing shutdown response time by 3%
- Designed a battery cover 3D CAD model and 3D printed it, resulting in a 50% lighter prototype

Undergraduate Researcher, *Queen Mary Centre for Undergraduate Research* Dec 2023 – Present

- Researching the topic titled "*Sway and Play: The Mathematics of Playground Swinging*" under the supervision of Dr. Rehan Shah which involves applying advanced mathematical techniques, such as Lagrangian mechanics and employing the use of Python
- Developing and implementing a Python toolkit to aid incoming engineering students with programming
- Presented research at the British Conference of Undergraduate Research (**BCUR**) 2024, *In Person* at the London School of Economics (**LSE**), and at Tomorrow's Mathematicians Today (**TMT**) conference 2024 and 2025, *Online*
- Authoring a related paper to publish within 4 months to an undergraduate journal

Tutor, *Learn and Grow Tuition* – Stoneleigh, Epsom Apr 2023 – Jul 2023

- Provided personalised tutoring for students in Years 3–11, supporting development of English and Mathematics
- Adapted lesson plans to accommodate diverse learning needs, including students with disabilities and varying aptitudes
- Coordinated with the organiser and fellow tutors to tailor instruction based on individual student progress and abilities

- Streamlined lesson preparation by ensuring all teaching materials and equipment were ready for use
- Operated industrial power tools to cut materials, ensuring safety and compliance for student handling
- Managed and maintained electronics, including serving as the sole operator of the department's 3D printer
- Delivered hands-on support by assisting students in practical lessons and facilitating classroom demonstrations

Projects

"Sway and Play: the Mathematics of Playground Swinging"

GitHub repository

- Incorporated advanced mathematical techniques, like Lagrangian mechanics, to produce and solve non-linear, non-homogeneous differential equations with time-varying coefficients through a numerical method, and a through python plots to corroborate findings against analytical solutions
- Aim to represent a model that is as close to lifelike as possible through the means of different strategies and incorporating friction
- *Tools used:* Python, MATLAB

Python Help Document

GitHub repository

- Co-created a Python help document, providing explanations and examples of common Python mistakes, implemented into the teaching curriculum to introduce newly beginning 1st year students to programming Python
- Obtained feedback from 190 students with overwhelmingly positive feedback, and 77% of respondents having 0 to 10 hours of programming experience
- Awarded the Certificate of best Intercalated/QMCUR Posters ILF 2025
- *Tools used:* LaTeX, Python

Investigation into the behaviour of fluid flow within a 90 degree pipe bend

Grade achieved: 94%

- Conducted group research into the fluid dynamics that occur within a pipe system
- Simulated meshes of various complexities on STARCCM+ to substantiate claims made in the report, allowing for an in-depth analysis into the flow phenomena in pipe bends
- *Tools used:* STARCCM+, Microsoft Office

Technologies and Languages

Languages: English (Proficient), Gujarati (Fluent in conversations)

Technical Languages: Python, HTML, CSS, JavaScript, C language

Software: Microsoft office applications, Microsoft Teams/Zoom, CREO, KiCad, RapidHarness, SolidWorks, VSCode, MATLAB, Simulink, AspenPlus, GITHUB, STARCCM+ +, Ansys Mechanical, Ansys Granta Edupack

Skills: Problem-solving, analytical research, public speaking, critical thinking, time management, teamwork, leadership